THE UNIVERSITY OF AUCKLAND

SECOND SEMESTER, 2020
Campus: City

MATHEMATICS

Topology

(Time allowed: TWO hours)

NOTE: SOLUTIONS

1. (8 marks)

F,F,T,T,T,F,F,T

2. (15 marks)

(a) $X/\sim$ is homeomorphic to $\mathbb{R}$. They may set up a quotient function or argue that neighbourhoods of $[0,1]$ as an equivalence class are homeomorphic to neighbourhoods of 0 in $\mathbb{R}$.

(b) Let $\mathbb{Z}$ have the discrete topology. Let E be the set of even integers and O the odd integers. Define open sets of a to be cofinite subsets of E and including a, and open sets of b to be cofinite subsets of O and including b. They can then check compact and Hausdorfff which is straightforward.

3. (8 marks)

(a) No. Choose a rational number $q \in (0,1)$. Then $(0,q) \times [0,1] \cap X$ and $(q,1) \times [0,1] \cap X$ are disjoint open sets separating X.

(b) No. $\{(0,q) \times [0,1] \cap X : q \in \mathbb{Q}\}$ is an open cover with no finite subcover.

(c) Yes. We know that there is a countable dense subset of irrationals in $\mathbb{R}$.

(d) Yes, it is a subspace of a Hausdorff space.

4. (14 marks)

(a) A couple of spirals that meet at appropriate intervals ("above" the intersection point) will do it.

(b)

$$H_q^{\Delta}(X) = \begin{cases} \mathbb{Z} & \text{if} \quad q = 0 \\ \mathbb{Z} & \text{if} \quad q = 1 \\ 0 & \text{if} \quad q > 1 \end{cases}$$

With appropriate working.